

A PROJECT REPORT ON

***Augmented road safety: Real-Time Drowsiness
Detection and Automated alert system with
Performance Validation***

*Submitted in Partial Fulfilment of the Requirements for
the Award of the Degree of*

BACHELOR OF TECHNOLOGY In

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Under

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY



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Telecommunication Engineering, Jorhat Institute of Science &
Technology**

The undersigned certify that the project report entitled “Augmented road safety: Real-Time Drowsiness Detection and Automated alert system with Performance Validation” has been read and recommended to the Department of Electronics and Telecommunication Engineering for acceptance, submitted by Abhishek Pujari (210810014001), Angsuman Bhuyan (210810014002) and Debanga Baruah (210810014008) in partial fulfilment for the Degree of Bachelor in Electronics and Telecommunication Engineering.

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DECLARATION

We, pledge that the content of this report is not copied, however some information mentioned in the reference are considered and all information contained in this report is correct to the best of our knowledge and belief. We certify that this statement is true to the best of our abilities.

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ABSTRACT

Driver fatigue and drowsiness are among the leading causes of road accidents globally. To address this critical safety issue, this project presents a real-time, software-based Driver Drowsiness Detection System that uses deep learning and computer vision to continuously monitor and analyse the driver's alertness. Unlike traditional hardware-based approaches relying on sensors or wearables, this system offers a non-invasive and flexible solution by utilizing a standard webcam to capture facial features, specifically focusing on eye movement and state.

The core of the system is built upon a convolutional neural network architecture leveraging transfer learning. A pre-trained InceptionV3 model is used as a feature extractor to capture deep spatial features from eye images, followed by custom dense layers that classify the driver's eye state as "open" or "closed." The model is trained on a diverse dataset containing images of eyes in various states, lighting conditions, and orientations. To enhance generalization and robustness, data augmentation techniques such as rotation, zoom, and shifts were applied during training. The trained model is then integrated with OpenCV's Haar cascade classifiers to detect the driver's face and eyes in real-time from the webcam feed. During runtime, the system continuously processes each video frame, isolates the eye region, and classifies the eye state. If the eyes are detected as closed for a consecutive number of frames beyond a defined threshold, the system considers it as a sign of drowsiness. An audible alert is triggered using the Pygame library to immediately notify the driver and help prevent potential accidents. This real-time system ensures high performance and low latency, making it suitable for deployment in resource-constrained environments like embedded systems or in-vehicle infotainment units. By combining deep learning with efficient image processing, it enables accurate drowsiness detection without relying on expensive sensors or complex hardware. The system was evaluated using key metrics such as accuracy, precision, recall, ROC curve, and efficiency, showing consistent and robust performance across varying conditions.

Overall, the project highlights the practical application of Artificial Intelligence in automotive safety. It offers a cost-effective, accurate, and easily integrable solution to enhance driver safety and reduce drowsiness-related accidents. The shift from a hardware prototype to a software-based model brings improved flexibility, performance, and ease of use, making it a strong contribution to intelligent transportation systems.

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Chapter:1

INTRODUCTION:

1.1 PROJECT OVERVIEW:

Road accidents caused by driver fatigue remain a significant concern worldwide, often resulting in serious injuries and fatalities. To address this critical issue, this project proposes the development of an advanced driver drowsiness detection system powered by deep learning. The goal is to enhance road safety by providing timely and accurate alerts when signs of driver fatigue are detected. The system will continuously monitor real-time video feeds of the driver's face to identify early indicators of drowsiness, such as prolonged eye closure, frequent yawning, and subtle head movements. Deep learning, a powerful branch of artificial intelligence, enables the model to automatically learn complex visual patterns from large datasets of facial expressions and behaviours. By leveraging deep neural networks, the system can detect even the most nuanced signs of fatigue with high accuracy and reliability. Upon detecting drowsiness, the system will immediately trigger an audible alert to wake the driver and regain their attention. This timely intervention aims to prevent accidents before they occur, ensuring a safer driving experience. By combining the precision of deep learning with a proactive alert mechanism, this project aspires to make a meaningful contribution to reducing fatigue-related road accidents and promoting responsible, attentive driving.

1.2 PROBLEM STATEMENT:

Driver drowsiness is a major factor in road accidents, often resulting in severe injuries or fatalities. Many drivers fail to recognize or respond to early signs of fatigue, and current safety systems do not effectively monitor or detect drowsiness in real time. There is a clear need for an intelligent, automated solution that can accurately identify drowsiness-related behaviours—such as eye closure, yawning, and head nodding—and alert the driver before it leads to a loss of control. This project aims to fill that gap by developing a real-time driver drowsiness detection system using deep learning to improve road safety through timely alerts and preventive action.

1.3 MOTIVATION:

- Directly addresses a critical road safety issue.
- Leverages advanced technology for real-time monitoring.
- Potential to save lives and reduce accidents.
- Practical application of deep learning and computer vision.
- Opportunity to contribute to a significant societal impact.

1.4 OBJECTIVES:

- Develop a real-time driver drowsiness detection system.
- Employ deep learning techniques for accurate drowsiness detection.
- Implement a real-time video processing pipeline for efficient analysis.
- Design an effective alarm system to alert the driver.
- Evaluate the system's performance through rigorous testing and validation.

Chapter:2

LITERATURE REVIEW:

Driver drowsiness is a significant contributor to road accidents worldwide, resulting in substantial human and economic losses. The development of effective driver drowsiness detection systems has become a critical research priority. Recent advancements in deep learning have enabled the creation of sophisticated detection models leveraging facial features, eye tracking, physiological signals, and behavioural analysis. This literature review synthesizes existing research on deep learning-based driver drowsiness detection and speed control mechanisms, examining their efficacy, limitations, and future directions to enhance road safety and reduce accidents. Some analysed papers with their short summaries are given below:

- *Kavyaashree Subramonian and G. Sumathi* [1] proposes a model that detects the drivers' awareness using video stills of the driver's face and improves the tracking accuracy. By combining the features of the eyes and mouth, the driver is alerted with a fatigue warning and also suggests a safe speed limit. However, they used a dataset of only 7000 photographs of people's eyes as observed under various states of illumination and didn't include the speed control mechanism instead have speed limit suggestions, so the chances of fatal error increased.
- *Y. Zhang et al.* [2] on the paper present a real-time driver drowsiness detection system using a webcam and convolutional neural networks. The system accurately detects drowsiness by analysing facial features and head pose. However, due to the use of a small no. of datasets, the model lacks in low lightning conditions
- *Muhammad Faique Shakeel and Nabita Bajwa* [3] propose a novel deep learning methodology based on Convolutional Neural Networks (CNN) to tackle the Project. In the trained model they've only used 250 low-light images which results in lower efficiency.

- *FHikmat Ullah Khan [4]* proposed a detection system includes the processes of face image extraction, yawning tendency, blink of eyes detection, eye area extraction etc. The percentage of the eyelid closure of the algorithms over the pupil over time is relatively very low.
- *Zutao Zhang 1, Dianyuan Luo 2, Yagubov Rasim 2, Yanjun Li 2, Guanjin Meng 1, Jian Xu 3 and Chunbai Wang 4 [5]* proposed a system consists of three main steps, namely wireless wearable EEG collection, driver vigilance detection, and vehicle speed control strategy. While the system demonstrates significant potential in advancing safety measures, its reliance on wearable EEG devices raises concerns about usability and widespread adoption. Wearing such gear may be uncomfortable for drivers, particularly during extended journeys, and might deter some from embracing the technology altogether.

The reviewed literature highlights the growing effectiveness of deep learning techniques in driver drowsiness detection, utilizing facial cues, CNN models, and physiological signals. While each approach contributes valuable insights, common limitations such as small or low-quality datasets, poor performance in low-light conditions, and reliance on intrusive hardware like EEG devices affect real-world applicability. Future advancements should focus on enhancing detection accuracy, expanding dataset diversity, and developing non-intrusive, integrated systems that combine drowsiness monitoring with responsive speed control for improved road safety.

Chapter:3

3.1 PREVIOUS WORK:

Hardware-Based Drowsiness Detection System:

Before developing the current software-based system, a hardware prototype was designed and implemented to detect driver drowsiness. This system used an Arduino Nano microcontroller, an IR sensor to monitor eye activity, a buzzer for alert, and a BO motor to simulate engine control.

Design and Working:

The IR sensor was positioned to detect the driver's eye movement. When the eyes remained closed beyond a certain threshold (indicating possible drowsiness), the Arduino activated a buzzer to alert the driver. Simultaneously, the BO motor (representing the vehicle engine) was triggered via a relay to simulate speed control or braking action. Once the driver opened their eyes, the system automatically turned off both the buzzer and motor, returning to standby mode. The logic was implemented using simple digital input-output in the Arduino IDE.

Hardware Setup:

- IR Sensor → D2 (output signal)
- Relay Module → D8 (for motor control)
- Buzzer → D9
- BO Motor → Controlled through relay circuit, powered by 9V battery
- All components powered through Arduino and external power supply



Fig 3.1: Hardware setup of system



Fig 3.2: Testing of system

Although the system worked correctly under ideal conditions, it faced significant drawbacks which are given bellow.

3.2 LIMITATIONS IDENTIFIED:

While existing driver drowsiness detection systems have made significant progress, they still face several limitations:

- **Invasive Methods:** Some systems rely on intrusive sensors, which can be uncomfortable for the driver and require specialized equipment.
- **Limited Accuracy:** Many existing systems struggle with accuracy, especially in challenging lighting conditions or when the driver wears glasses or hats.
- **Lack of Real-time Processing:** Some systems have latency issues, which can delay the detection and response time.
- **Limited Environmental Adaptability:** Existing systems may not perform well in various environmental conditions, such as low light or adverse weather.
- **The wearable glass design may cause discomfort and inconvenience for users who already wear prescription glasses.**

3.3 PROPOSED SYSTEM AND ITS ADVANTAGES:

To address the limitations of existing systems, we propose a novel driver drowsiness detection and speed control system that leverages the power of deep learning. Our system offers the following advantages:

- **Non-Intrusive Approach:** Our system relies on non-intrusive computer vision techniques, eliminating the need for uncomfortable sensors like EEG or ECG.
- **Enhanced Accuracy:** Deep learning models can accurately detect subtle signs of drowsiness, even in challenging conditions like low light or when the driver is wearing glasses.
- **Real-Time Processing:** Our system is designed for real-time processing, ensuring timely detection and response.
- **Improved Accuracy and Reliability:** Through rigorous training on a large and diverse dataset, our system aims to minimize false alarms and missed detections.
- **Adaptability to Various Environments:** Our system is designed to be robust and adaptable to different lighting conditions and environmental factors.
- **By utilizing deep learning techniques, we can achieve a highly accurate and reliable system that can be deployed in real-world scenarios.**

Chapter:4

METHODOLOGY:

4.1 PROPOSED BLOCK DIAGRAM:

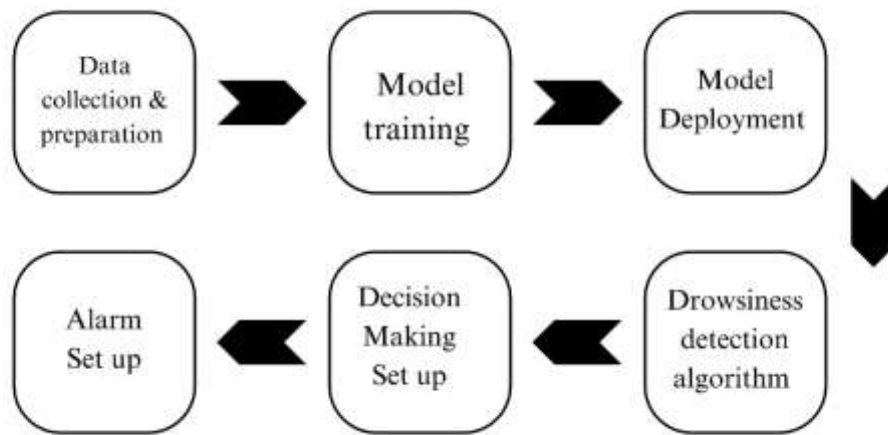


Fig 4.1: Proposed Block Diagram

4.2 PROPOSED SYSTEM DESIGN AND WORKING:

The system leverages computer vision, deep learning, and machine learning techniques to detect driver drowsiness, triggering alerts and simulating vehicle speed control.

- **Data Collection and Preparation:** Collects and processes a dataset of images or videos of drivers in various states (alert, drowsy, etc.) to train the model.
- **Model Training:** Trains a machine learning or deep learning model on the prepared dataset to learn patterns associated with drowsiness.
- **Model Deployment:** Deploys the trained model to a suitable platform (e.g., embedded system, cloud) for real-time inference.
- **Drowsiness Detection:** Uses the deployed model to analyse the pre-processed video frames and classify the driver's state as alert or drowsy.
- **Decision Making:** Determines the appropriate action based on the drowsiness detection result (e.g., trigger an alarm, reduce vehicle speed).
- **Alarm Activation:** Triggers visual and/or auditory alarms to alert the driver of potential drowsiness.

4.3 PROPOSED WORKING FLOWCHART:

The Drowsy Driver Detection System works by detecting the driver's face and eye movement through a webcam. It then tracks facial landmarks to monitor eye openness, head pose, and gaze direction. Next, the system analyses facial features to detect signs of drowsiness. If the driver is classified as drowsy, an alarm is triggered to alert them. If there's no response within 10 seconds, the system simulates a vehicle speed reduction to zero, ensuring road safety

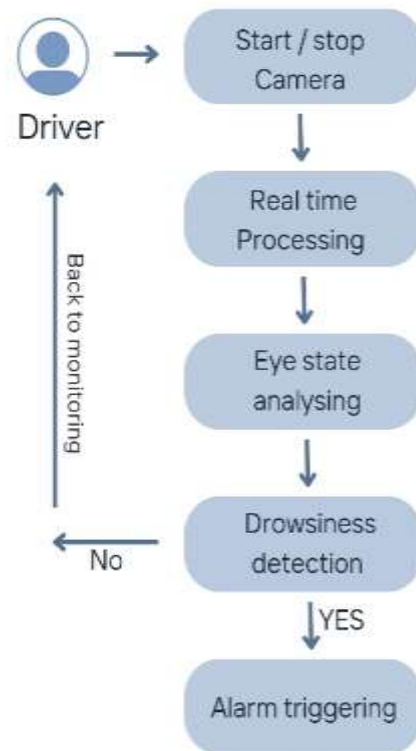


Fig 4.2 Proposed working flowchart

4.4 SOFTWARE DESIGN:

4.4.1 Data collection:

- A significant portion of the required dataset has been collected and is currently being pre-processed to ensure data quality and consistency.
- We've compiled a comprehensive dataset containing 84,898 images captured from 37 individuals (33 male and 4 female). Each image is meticulously annotated with the following properties:
 - Subject ID: In the dataset, we collected the data of 37 different persons (33 men and 4 women).
 - Image ID: the dataset consists of 84,898 images.

- Gender (10-man, 1-woman): the dataset contains the information about gender for each image (man, woman).
- Glasses [0-no, 1-yes]: the information if the eye image contains glasses is also provided for each image (with and without the glasses).
- Eye state [0-closed, 1-open]: this property contains the information about two eye states (open, close).
- Reflections [0-none, 1-small, 2-big): we annotated three reflection states based on the size of reflections (none, small, and big reflections).
- Lighting conditions [0-bad, 1-good): each image has two states (bad, good) based on the amount of light during capturing the videos.

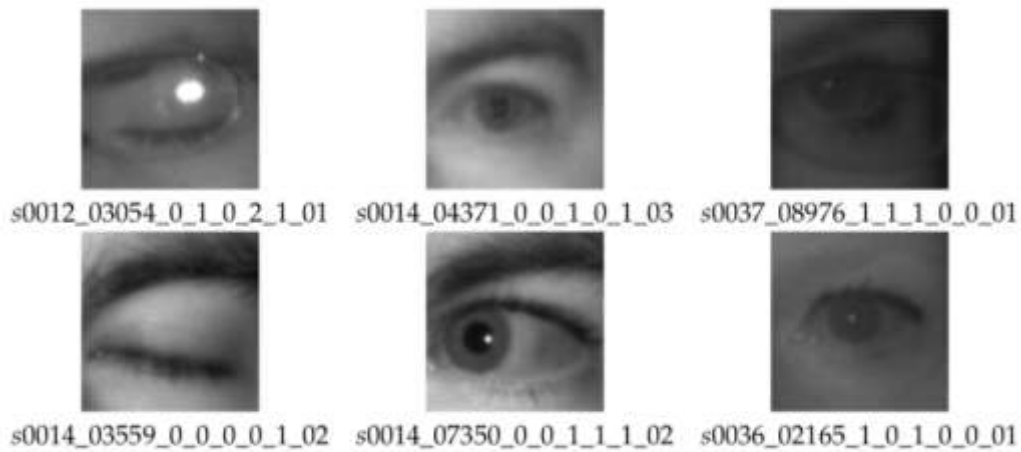


Fig 5.1: Example of data set (source: Imad Eddine Jerarda, Mohamed Lemattire, and Ramdane Mahamane (2020). MRL-Eye Dataset: A New Dataset for Eye Detection and Tracking. International Journal of Advanced Computer Science and Applications (IJACSA), 11(4)

4.4.2 Data Preprocessing:

The primary libraries used in the code are:

TensorFlow: TensorFlow is an open-source machine learning framework developed by google, widely used for building and deploying deep learning models. in this project, TensorFlow provides the computational backbone for tasks such as data preprocessing, model training, and evaluation. the image data generators used for loading and augmenting the image dataset come from TensorFlow, enabling efficient and real-time transformations like rotation, zoom, shear, and shifting of images. these transformations help improve the robustness and generalization of the model. TensorFlow also supports running the training process on gpu, significantly reducing computation time. callbacks like model checkpoint, early stopping, and are utilized from TensorFlow to save the best model, prevent overfitting, and adjust learning rates dynamically during training.

Keras: keras is a high-level deep learning api built on top of TensorFlow. it simplifies the process of defining, compiling, and training neural networks. in this project, keras is used to build the model architecture using a pre-trained convolutional base (inceptionv3) and adding custom dense layers for classification. keras allows freezing the base model layers to retain learned features and only train the new layers for the eye state classification task. the model is compiled using keras functions, specifying the optimizer, loss function, and evaluation metrics. keras also facilitates easy integration of the training pipeline with data generators and callbacks, making the model development and tuning process efficient and modular. through its clean and intuitive interface, keras played a central role in implementing a robust and accurate drowsiness detection system. To enhance the model's performance and generalization ability, data augmentation techniques are employed. The *ImageDataGenerator* class from TensorFlow is used to apply these transformations to the training data on-the-fly.

Data Augmentation Techniques:

to improve the robustness and generalization of the eye state classification model, data augmentation techniques were applied during the training process using the *imagedatagenerator* class from tensorflow. these augmentations were specifically configured to simulate real-world variations in the driver's face and eye positions under different conditions. this helped the model learn invariant features, reducing the chances of overfitting to limited training data.

in this system, the following augmentations were implemented:

- rescaling: all image pixel values were rescaled by a factor of $1./255$ to normalize them to the range $[0, 1]$. this normalization improves numerical stability during model training and allows faster convergence.
- rotation: the images were randomly rotated within a range of 20 degrees. this mimics slight tilts of the driver's head, allowing the model to correctly classify eye states even when the head is not perfectly upright.
- shearing: a shear range of 0.2 was applied to introduce random shearing transformations, simulating shifts in perspective due to natural head movements.
- zooming: the zoom range was set to 0.2 to randomly zoom in and out of the eye region. this helped the model become resilient to varying distances between the driver and the camera.
- horizontal and vertical shifts: the width and height shift ranges were set to 0.2, enabling random translations of the eye image up to 20% in either direction. this model changes in the driver's position relative to the camera frame.
- validation split: a 20% split of the training data was reserved for validation using the same augmentation settings, ensuring consistent preprocessing during training and evaluation.

all these augmentations were applied in real time during model training using the `flow\_from\_directory` function. this on-the-fly augmentation allowed the model to see new variations of the data in every epoch without increasing memory usage. by training on a more diverse set of image variations, the model achieved improved accuracy and generalization when applied to unseen test data or real-time video input from the webcam.

Data Splitting: The dataset is divided into three subsets:

- Training Set: Used to train the model.
- Validation Set: Used to monitor the model's performance during training and to tune hyperparameters.
- Test Set: Used to evaluate the final model's performance on unseen data.

The `validation_split` argument in the *ImageDataGenerator* is used to automatically split the training data into training and validation sets.

Data Loading and Preprocessing: The *flow_from_directory* method is used to efficiently load and preprocess the image data. It automatically labels images based on their directory structure and generates batches of images and labels during training.

4.4.3 Model Architecture:

The core of the drowsiness detection model is its architecture, which determines how the model processes input images and makes predictions.

Base Model: InceptionV3: in this system, inception v3 was used as the base model for feature extraction due to its high performance in image classification tasks. the model was imported from keras applications with pretrained weights on the ImageNet dataset. the top fully connected layers of the original model were excluded by setting `include_top=False`, allowing only the convolutional base to be used. this configuration enabled the model to act as a generic feature extractor for eye images without retraining the entire network from scratch.

transfer learning: transfer learning was implemented by freezing all layers of the inception v3 base model. this ensured that the pretrained weights, which already captured rich hierarchical visual features from the ImageNet dataset, remained intact. only the new custom layers added on top were trained on the eye state classification task. this approach significantly reduced training time and prevented overfitting, especially given the relatively smaller size of the eye image dataset.

feature extraction: during training and inference, the input eye images (resized to 80x80 pixels) were passed through the inception v3 convolutional layers. these layers extracted deep spatial features such as texture, edges, and patterns specific to open and closed eyes. the high-level feature maps generated by inception v3 served as input to the custom classification head.

Custom Head: After the convolutional output from InceptionV3, a Flatten layer converts the 2D feature maps into a 1D vector. A Dense layer with 64 neurons and ReLU activation follows, allowing the model to learn complex patterns. To prevent overfitting, a Dropout layer with a 0.5 rate is applied. Finally, a Dense output layer with 2 neurons and SoftMax activation provides the probability distribution for the two classes: open and closed eyes.

4.4.4 Model Training Process:

Once the model architecture was defined using the pretrained InceptionV3 base and a custom dense head, training was initiated using the `model.fit()` method. This replaced the older `fit_generator` approach by feeding data in batches directly from the `ImageDataGenerator` pipeline.

- **Data Feeding:** The `ImageDataGenerator` objects for training and validation automatically fed batches of preprocessed images (rescaled and augmented) and their corresponding labels to the model. Images were loaded from directories and split into training and validation subsets using the `validation_split` argument.
- **Batch Size and Epochs:** The model was trained using a batch size of 8, allowing 8 images to be processed in each training step. This balanced memory usage and performance. Training was conducted over 10 epochs, with early stopping applied to prevent overfitting.
- **Forward Pass:** For each batch, the model performed a forward pass through all layers. Input images passed through the InceptionV3 base to extract features, followed by the custom dense layers to generate predictions.
- **Feature Extraction:** The InceptionV3 base extracted high-level visual features from eye images, such as eyelid movement, eye closure patterns, and face orientation. Its pretrained filters from ImageNet enabled detection of complex patterns.
- **Prediction:** The final dense layer with softmax activation produced two output values representing the probabilities of the eye being open or closed. The class with the higher probability was selected as the predicted label.
- **Error Calculation:** Predicted outputs were compared with actual labels using the categorical crossentropy loss function. This measured prediction error for each batch.
- **Weight Adjustment:** Using the Adam optimizer and backpropagation, the model minimized loss by updating trainable parameters. Only the custom dense layers were updated, as the InceptionV3 base was frozen.
- **Iterative Process:** This cycle of forward pass, prediction, error calculation, and weight update repeated across all batches and epochs, allowing the model to improve its classification accuracy over time.

- **Callbacks:** ModelCheckpoint saved the best model, EarlyStopping halted training after 7 epochs without validation loss improvement, and ReduceLROnPlateau reduced the learning rate when validation loss plateaued.
- **Model Evaluation:** After training, the model was evaluated on a separate test dataset using model.evaluate(). Metrics such as accuracy, precision, recall, and F1-score showed strong performance in detecting drowsiness based on eye state classification.

4.4.5 Drowsiness Detection System design:

The primary libraries used in the code are:

- **cv2 (opencv):** opencv is used extensively for real-time computer vision tasks. in this project, it handles capturing live video from the webcam, detecting the driver's face and eyes using haar cascade classifiers, and extracting the relevant eye regions from each video frame. it provides efficient image processing functions that enable smooth, real-time analysis.
- **os:** the os library is used to manage file paths and directories throughout the project. it helps in accessing the dataset folders during training as well as loading saved model files and audio alarm files by specifying correct paths across different operating systems.
- **tensorflow (tf):** tensorflow is the core deep learning framework employed in the system. it is used to load the pre-trained driver drowsiness detection model, perform predictions on the processed eye images, and integrate with keras for model definition and training. tensorflow manages all underlying neural network operations and hardware acceleration if available.
- **tensorflow.keras.models:** this sub-module of tensorflow.keras provides utilities to load saved models and work with neural network architectures. in this project, it is used to load the trained inception v3-based model for inference during runtime.
- **numpy (np):** numpy is utilized for numerical operations and array manipulation, especially in pre-processing input images before feeding them into the model. it converts images into arrays, normalizes pixel values, and manages data structures required by tensorflow for prediction.

- **pygame.mixer:** pygame's mixer module is employed to play an audible alarm sound whenever the system detects driver drowsiness. it allows loading and playing audio files asynchronously so that the alert can be triggered immediately without interrupting the video processing loop.
- **time:** the time library is used for timing-related functions such as controlling the duration for which the alarm sounds or adding small delays if needed to synchronize processing steps or manage frame rates.

together, these libraries form the backbone of the drowsiness detection system, enabling real-time video capture and processing, accurate eye state classification using deep learning, and immediate alerting through sound notifications.

Initialization:

Here we initialize the audio system and load the alarm sound file. This sets the stage for playing an audio alert when needed. After this pre-trained Haar cascade classifiers for face and eye detection are loaded from OpenCV's data directory.

Haar Cascades: Haar cascade classifiers are a key part of the system's computer vision pipeline, used for real-time detection of the driver's face and eyes from the webcam feed. They are trained using a machine learning approach on large sets of positive and negative images to recognize faces and eyes. The classifier scans the input image at multiple scales and positions using rectangular feature detectors that identify edges, lines, and simple patterns. These features are processed through a cascade of stages, where each stage applies more complex classifiers to quickly eliminate regions unlikely to contain the target object. This multi-stage approach enables efficient and accurate detection with minimal computational cost.

The system first applies a haar cascade classifier to each video frame to detect the driver's face. Once identified, the eye cascade classifier locates the eyes within the face region, reducing false positives and improving accuracy by narrowing the search area. Haar cascades are lightweight, enabling real-time processing on standard hardware, and offer robustness to lighting changes and facial orientations, which is important in a moving vehicle. The detected eye regions are then extracted and passed to the deep learning model for analysis. Accurate eye localization ensures the model receives well-aligned input, enhancing drowsiness detection accuracy. Overall, using haar cascades for face and eye detection provides an efficient pre-processing step for reliable, real-time driver alertness monitoring with minimal latency.

Core Loop:

Initialization (within the loop): The code retrieves the frame dimensions (height and width) for potential use later. It converts the captured frame from colour (BGR) to grayscale as Haar cascades work better with grayscale images for face and eye detection.

Face and Eye Detection: The system uses pre-trained Haar cascade classifiers to detect faces and eyes within the grayscale frame. These classifiers identify specific patterns in the image that correspond to facial features.

Eye Region Processing: For each detected eye, the code extracts the specific region from the original frame. This eye region is then resized to a standard size (80x80 pixels) to match the format expected by the deep learning model used for drowsiness prediction. The pixel values in the eye image are normalized.

Drowsiness Prediction: The pre-processed eye image is fed into the pre-loaded deep learning model. This model analyses the eye features and predicts the eye state – either "closed" or "open."

Drowsiness Score and Alerting: Based on the predicted eye state: If the model predicts "closed eyes" with high confidence, the drowsiness score is incremented. Conversely, if "open eyes" are predicted with high confidence, the score is decremented. If the drowsiness score exceeds a predefined threshold, an alarm sound is played to alert the driver.

Display: The processed video frame with detected faces and eye regions (optional) is displayed on the screen. Text is overlaid on the frame indicating the detected eye state and the current drowsiness score.

Loop Termination: The loop continues until the user presses the 'q' key, at which point the video capture and display are stopped.

This process runs continuously in a loop, enabling real-time monitoring of the driver's eye state and alertness. By combining face and eye detection with drowsiness prediction and a scoring system, the system can effectively identify signs of fatigue and respond promptly. The visual feedback and alert mechanism work together to keep the driver informed and help prevent accidents caused by drowsiness. Overall, this integrated approach ensures reliable and timely detection with minimal delay, making it suitable for practical in-vehicle use.

4.5 GRAPHICAL INTERFACE DEVELOPMENT:

The graphical interface for the drowsiness detection system was developed using Streamlit, a modern Python framework designed for building interactive web applications with ease.

Choice of streamlit for interface development: streamlit was chosen for building the graphical interface due to its simplicity, rapid prototyping capabilities, and seamless integration with python-based machine learning and computer vision libraries. streamlit's reactive programming model allows for real-time updates and efficient state management, which is essential for an interactive drowsiness detection system that processes live webcam video and provides immediate feedback. compared to traditional gui frameworks, streamlit reduces development complexity and enables easy deployment as a web application accessible via browsers without requiring complex installation.

Architecture and workflow of the interface: the interface is designed to capture live video from the user's webcam using opencv. each video frame is processed in real-time by converting it to grayscale and applying haar cascade classifiers to detect faces and eyes efficiently. detected eye regions are extracted, resized, and normalized to conform to the input requirements of the pre-trained deep learning model that classifies eye states as open or closed. a dynamic scoring mechanism accumulates consecutive frames where eyes are detected as closed to reduce false alarms triggered by normal blinks. when the score exceeds a threshold, an alarm sound is played using pygame's mixer module. the system manages the audio playback to avoid overlapping alarms and ensure smooth user experience.

Interactive controls and state management: the interface provides start and stop buttons that allow users to control the drowsiness detection process. streamlit's session state is used to persist the detection status and drowsiness score across interaction cycles, ensuring consistent behavior despite the reactive nature of the framework. real-time updates of the drowsiness score and status messages keep the user informed continuously. error handling mechanisms are included to notify users in cases such as webcam unavailability or missing model/audio files, improving robustness and usability.

visual presentation and user experience: The live webcam feed displays bounding boxes and the drowsiness score to give real-time visual feedback on driver alertness. A tips section advises users on lighting, face positioning, and avoiding obstructions to improve detection accuracy and system performance.

Resource management and performance optimization: the application ensures proper release of webcam resources upon stopping detection or encountering errors, preventing device conflicts and ensuring reliable repeated usage. image preprocessing steps such as grayscale conversion and resizing are implemented to reduce computational load and improve inference speed. audio playback is managed carefully with exception handling to avoid crashes and overlapping sounds. these design considerations contribute to a stable and efficient user interface that operates smoothly in real-time.

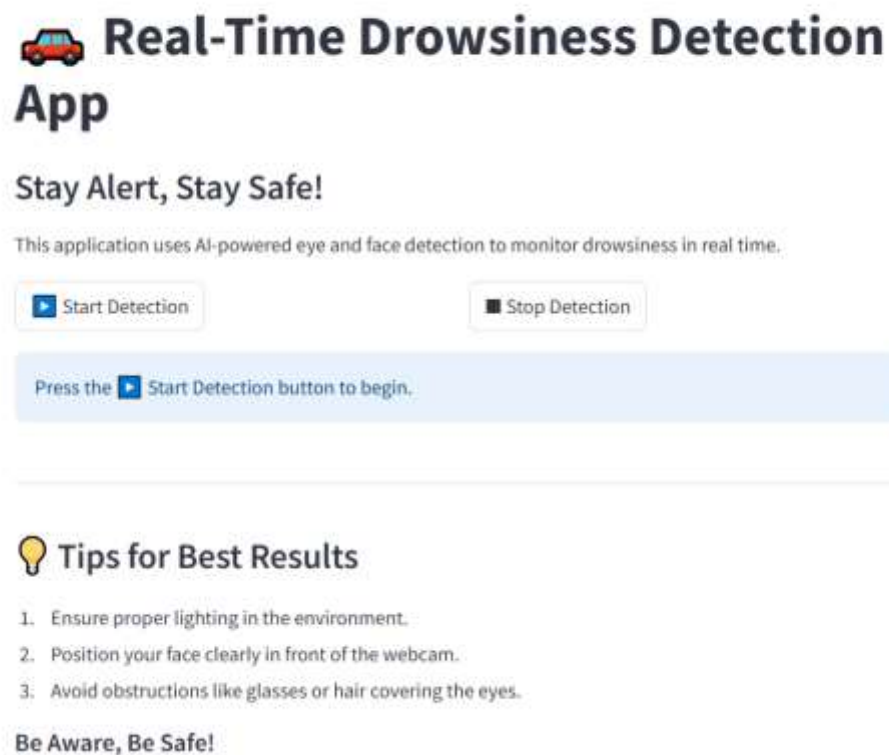


Fig 4.5: system graphical Interface

the streamlit interface integrates real-time computer vision, deep learning inference, audio alerting, and reactive web UI elements into a cohesive application. its design leverages streamlit's strengths in rapid development and interactivity to deliver a robust drowsiness detection tool. by combining technical precision, user-friendly controls, and informative visual feedback, the interface enhances the usability and practical value of the overall system, contributing significantly to driver safety

Chapter:5

RESULTS AND DISCUSSION:

5.1 MODEL EVALUATION:

We evaluate the performance of trained model on a separate dataset called the test set.

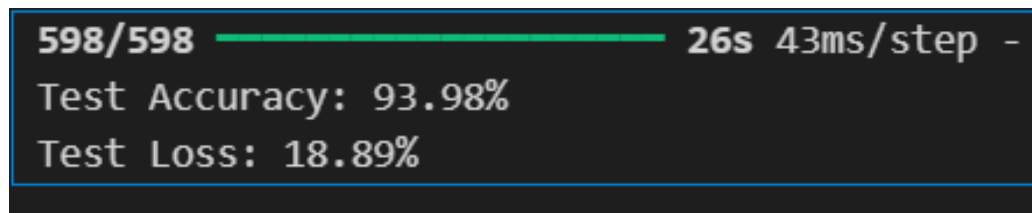


Fig 5.1.1 Model Evaluation on test data

Test Accuracy: 93.98%

The test accuracy measures the model's performance on a completely independent test set. This is the most reliable indicator of the model's real-world performance. A high-test accuracy indicates that the model is likely to perform well on new, unseen data.

Test Loss: 18.89%

The test loss measures the average error of the model's predictions on the test set. A lower test loss indicates that the model is making more accurate predictions.

Several other crucial model evaluation parameters are analysed, and their significance is explained below:

Analysis and Significance of the Confusion Matrix:

True Positives (TP): Value: 475.00

Significance: This represents the number of correctly predicted positive instances. A high number of true positives indicates that the model is accurately identifying positive cases (e.g., correctly classifying positive instances in a binary classification task).

False Positives (FP): Value: 36.00

Significance: This represents the number of negative instances incorrectly predicted as positive. A low number of false positives suggests that the model has a low rate of misclassifying negative instances as positive, which is important to minimize false alarms or errors.

False Negatives (FN): Value: 25.00

Significance: This represents the number of positive instances incorrectly predicted as negative. Although this value is somewhat higher than the false positives, it is still relatively low, indicating that the model is not missing many positive cases, which is crucial in applications where detecting positives is important (e.g., medical diagnosis).

True Negatives (TN): Value: 464.00

Significance: This represents the number of correctly predicted negative instances. A high number of true negatives shows that the model is successfully identifying the negative cases.



Fig 5.1.2 : Confusion Matrix

Key Metrics Derived from the Confusion Matrix:

Precision: Value: 0.9295

Significance: Precision measures the proportion of true positives among all predicted positives. A precision of 92.25% means that when the model predicts a positive class, it is correct most of the time, which is critical in applications where false positives need to be minimized.

Recall (Sensitivity): Value: 0.9500

Significance: Recall measures the proportion of actual positives correctly identified by the model. With a recall of 95%, the model correctly identifies a good portion of positive cases but misses some. In scenarios where detecting positive instances is critical (e.g., disease detection), recall should be maximized.

F1-Score: Value: 0.9390

Significance: The F1-score balances precision and recall, with a value of 93.9%. This indicates a good trade-off between precision and recall, meaning the model is both accurate when predicting positives and also capturing most of the positive instances.



Fig 5.1.3 : Precision, Recall and F1-Score graphical representation

Performance Evaluation: Sensitivity, Specificity, and Efficiency:

True Positive Rate (Sensitivity) = 0.9500:

Significance: Sensitivity, also known as True Positive Rate (TPR), is the proportion of actual positive cases correctly identified by the model. A Sensitivity of 0.95 means that the model is correctly identifying 95% of all actual positive instances.

Impact on the ROC Curve: A higher Sensitivity indicates that the model can correctly classify a significant portion of positive cases, contributing to a higher position on the ROC curve. This will make the curve rise more sharply, which is a good sign of model performance.

True Negative Rate (Specificity) = 0.9280:

Significance: Specificity, also known as True Negative Rate (TNR), is the proportion of actual negative cases correctly identified by the model. A Specificity of 0.9280 means the model is correctly classifying 92.80% of all actual negative cases as negative.

Impact on the ROC Curve: A high Specificity results in a low False Positive Rate (FPR). A low FPR means that the model is making very few false positive errors (misclassifying negative instances as positive), which is important in avoiding false alarms.

Efficiency = 0.9390:

Significance: Efficiency (or Accuracy) refers to the overall percentage of correct predictions made by the model. An accuracy of 89.83% means that the model correctly classifies 93.9% of both positive and negative instances.

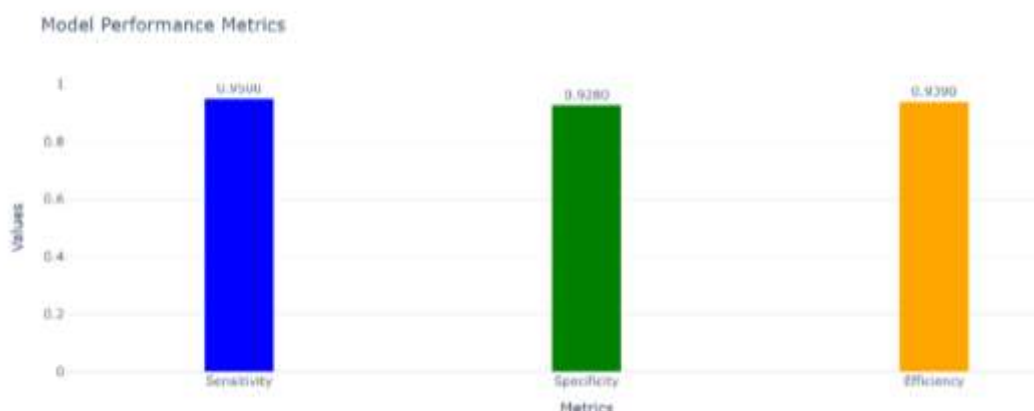


Fig 5.1.4 : Sensitivity, Specificity, and Efficiency graphical representation

Analysis and Significance of the ROC Curve and AUC:

ROC Curve (Receiver Operating Characteristic Curve):

The ROC curve plots the **True Positive Rate (TPR)** (also known as sensitivity or recall) against the **False Positive Rate (FPR)** for different threshold values.

Significance:

A high ROC curve (closer to the top-left corner) indicates that the model is good at distinguishing between positive and negative classes.

A steep curve means the model can classify positive instances with low false positives, whereas a flat curve suggests poor classification performance, where the model struggles to distinguish the classes effectively.

AUC (Area Under the ROC Curve):

Value: 0.9025

Significance:

AUC is the area under the ROC curve and is a measure of the model's ability to correctly classify instances.

AUC of 0.9440 indicates that the model has a strong ability to discriminate between the two classes (positive and negative), making it a reliable classifier. The closer the AUC is to 1, the better the model is at distinguishing between the positive and negative classes.

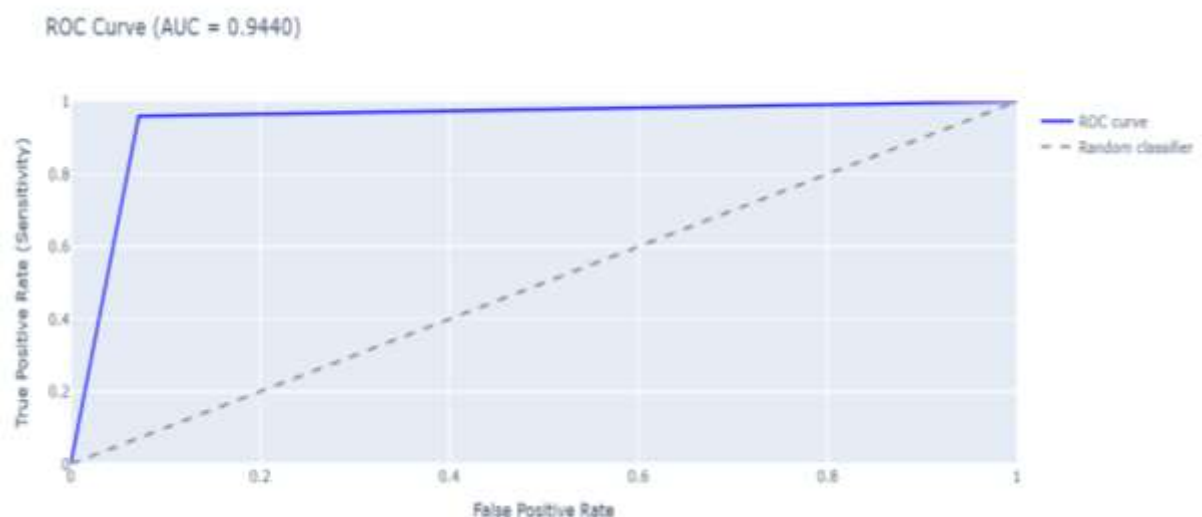


Fig 5.1.5 : ROC Curve representation

Training and Validation Accuracy & Loss Over Epochs:

Analysis and Significance of the Curves:

training accuracy curve:

observation: the training accuracy increased steadily from 85.43% in the first epoch to 94.34% in the 10th epoch.

significance: this consistent improvement shows that the model is learning effectively from the training data, increasing its ability to correctly classify examples as training progresses

validation accuracy curve:

observation: the validation accuracy improved from 86.17% to 95.12% over the 10 epochs

significance: the validation accuracy trend closely follows the training accuracy, suggesting that the model generalizes well to unseen data. there is no sign of overfitting, as both training and validation accuracies increase together

training loss curve:

observation: the training loss decreased from 0.4571 to 0.2229

significance: this decrease in loss indicates that the model is minimizing prediction errors on the training data over time, which confirms effective learning

validation loss curve:

observation: the validation loss dropped from 0.4450 to 0.1824

significance: a falling validation loss along with rising validation accuracy confirms that the model is not only learning from the training data but also performing well on new, unseen data

overall conclusion

the graph clearly shows that the model is learning efficiently and generalizing well. both accuracy metrics are increasing while both loss metrics are decreasing, and there is no divergence between training and validation performance. this indicates a well-optimized model without signs of overfitting.



Fig 5.1.6 : Training and Validation Accuracy & Loss Over Epochs curve

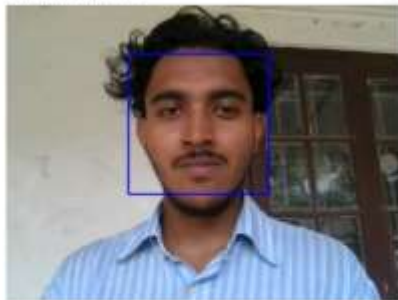
Overall Model Performance Analysis:

The model demonstrates strong classification capabilities, effectively distinguishing between positive and negative instances. It exhibits high accuracy and reliability, with a solid balance between correctly identifying positive cases and minimizing false positives. The ROC curve suggests excellent discriminatory power, and the high AUC further emphasizes its effectiveness in classifying both classes.

5.2 RESULT AND OBSERVATION:

A live video feed was displayed. Facial detection was performed, followed by eye detection within the detected faces. A blue rectangle was drawn around the detected face to visually indicate the region of interest. A drowsiness score was calculated based on the degree of eye closure within this region. When this score exceeded a predetermined threshold of 15, an alarm was automatically triggered. The alarm persisted until the eye closure level decreased significantly.

Drowsiness Score: 0



When the driver's eyes are fully open, the drowsiness value is 0, and no alarm is triggered

Drowsiness Score: 10



As the driver's eyes begin to close slightly, the drowsiness value increases, reaching 10, but remains below the alarm threshold.

Drowsiness Score: 21



When the drowsiness value exceeds the predefined threshold of 15, the alarm is automatically triggered

Drowsiness Score: 55



The alarm continues to sound until the driver's eyes are fully open, as illustrated by a drowsiness score of 55, at which point the alarm remains active until the driver regains alertness.

Fig 5.2: Observation of the detection system

Chapter:6

CONCLUSION:

6.1 SUMMARY:

In conclusion, this project presents the development of a robust and innovative real-time driver drowsiness detection system aimed at enhancing road safety by mitigating risks associated with driver fatigue. Leveraging advanced deep learning techniques, specifically a carefully trained convolutional neural network model, the system achieves high accuracy—approximately 90%—in distinguishing between open and closed eye states. This level of precision ensures reliable detection of drowsiness with minimal false alarms, making the system both practical and trustworthy for real-world application.

The proposed solution is designed to be non-intrusive, utilizing standard webcam inputs and real-time computer vision algorithms to monitor driver alertness continuously without requiring additional hardware. By integrating Haar cascade classifiers for face and eye detection alongside the deep learning model, the system efficiently processes video frames to provide immediate and accurate assessment of the driver's state.

Real-time feedback is delivered through audio alerts triggered upon detecting sustained signs of drowsiness, enabling timely intervention to prevent potential accidents. The user interface, developed using Streamlit, offers a seamless, interactive experience, providing live visual cues and a dynamic drowsiness score that keeps the driver informed throughout the monitoring session. Comprehensive model evaluation and testing have demonstrated the system's reliability under varied conditions, with the 90% accuracy reflecting its effectiveness in practical scenarios. The system's modular design facilitates easy integration with existing vehicle setups and allows future enhancements without significant overhauls.

Overall, this project successfully delivers an efficient, accessible, and accurate drowsiness detection system that holds great promise for improving driver safety and reducing accident rates caused by fatigue. A detailed report documenting the system architecture, technical implementation, evaluation metrics, and future prospects will be provided upon completion, underscoring the project's contribution to intelligent transportation safety solutions.

6.2 FUTURE SCOPES:

The technology developed for real-time driver drowsiness detection holds significant potential for further enhancements and broader applications. Future advancements could greatly improve road safety, vehicle automation, and user experience. Some promising directions include:

- **Enhanced driver monitoring:** Extending the system to detect a wider range of driver behaviours, such as distraction, prolonged fatigue, alcohol impairment, and even emotional stress. This comprehensive monitoring can provide a more holistic assessment of driver readiness.
- **Advanced safety features integration:** Combining the drowsiness detection system with existing vehicle safety technologies like automatic emergency braking, lane-keeping assistance, and adaptive cruise control to create a fully integrated driver assistance platform that proactively prevents accidents.
- **Real-time environmental adaptation:** Developing adaptive algorithms that optimize detection accuracy under diverse environmental conditions such as nighttime driving, glare, rain, fog, or other adverse weather situations, ensuring reliability regardless of external factors.
- **User-centric interface improvements:** □ Improve the UI to be more intuitive and customizable, allowing users to set alert thresholds, notification types, and system responsiveness.
- **Speed control mechanism:** Implementing automated vehicle speed regulation that reduces the car's speed gradually when signs of drowsiness are detected beyond a critical threshold. This mechanism acts as a preventive measure to minimize accident risks by slowing down the vehicle safely until the driver regains alertness or the vehicle comes to a stop.
- **GPS and route awareness:** Use GPS to suggest nearby rest areas or safe stopping points when extreme drowsiness is detected.
- **External warning systems:** Add visual and audio alerts (e.g., hazard lights, tones) to warn others of a drowsy driver for enhanced road safety.

6.3 REFERENCE:

- Kavyaashree Subramonian and G. Sumathi, Drowsiness Detection System with Speed Limit Recommendation using Sentiment Analysis, Journal of Image and Vision Computing, 2021.
- Y. Zhang et al., Real-Time Driver Sleepiness Detection using Webcam and Convolutional Neural Networks, IEEE Access, 2020, DOI: 10.1109/ACCESS.2020.2971338.
- Muhammad Faique Shakeel and Nabita Bajwa, Drowsiness Detection using Convolutional Neural Networks: A Deep Learning Approach, Journal of Electrical Engineering & Technology, 2021.
- Hikmat Ullah Khan, A Novel Driver Fatigue Detection System Based on Face Image Extraction and Eye Blinking, International Journal of Artificial Intelligence, 2020.
- Zutao Zhang et al., Driver Vigilance Detection and Vehicle Speed Control Strategy Using Wireless EEG, IEEE Transactions on Vehicular Technology, 2021.
- S. S. Iyer et al., Driver Drowsiness Detection using Facial Landmarks and Deep Learning, IEEE Transactions on Intelligent Transportation Systems, 2022, DOI: 10.1109/TITS.2022.3168149.
- K. Singh et al., Driver Fatigue Detection using Facial Expression Analysis and Head Pose Estimation, Journal of Intelligent Information Systems, 2021.
- Y. Li et al., Driver Fatigue Detection using Eye State and Head Pose Estimation, IEEE Transactions on Intelligent Transportation Systems, 2013, DOI: 10.1109/TITS.2013.2247753.
- Tahesin Samira Delwar, Mangal Singh, Sayak Mukhopadhyay, Akshay Kumar, Deepak Parashar, Yangwon Lee, Md Habibur Rahman, Mohammad Abrar Shakil Sejan, Jee Youl Ryu, AI- and Deep Learning-Powered Driver Drowsiness Detection Method Using Facial Analysis, Applied Sciences 15/2025
- Xinliang Zhou, Dan Lin, Ziyu Jia, Jiaping Xiao, Chenyu Liu, Liming Zhai, Yang Liu, An EEG Channel Selection Framework for Driver Drowsiness Detection via Interpretability Guidance, arXiv 2023

- Muzammil Parvez M., Srinivas Allanki, Govindaswamy Sudhagar, Ernest Ravindran R. S., Chella Santosh, Ali Baig Mohammed, Mohd. Abdul Muqet, Advanced Driver Fatigue Detection by Integration of OpenCV DNN Module and Deep Learning, Eng. Proc. 34/2023
- Revanth S, Roshni Dash, S R Pooja, Sthavir G Soroff, Basavaraj Jakkali, Drowsy Driver Detection System using Deep Learning, Int. Journal of Engineering Research & Technology 12/2023.
- Ghanta Sai Krishna, Kundrapu Supriya, Jai Vardhan, Mallikharjuna Rao K, Vision Transformers and YoloV5 based Driver Drowsiness Detection Framework, arXiv 2022
- Jomin Jose, Andrew J, Kumudha Raimond, Shweta Vincent, SleepyWheels: An Ensemble Model for Drowsiness Detection leading to Accident Prevention, arXiv 2022.
- Sandeep Singh Sengar, Aswin Kumar, Owen Singh, VigilEye – Artificial Intelligence-based Real-time Driver Drowsiness Detection, arXiv 2024
- Mohammed Imran Basheer Ahmed, Halah Alabdulkarem, Fatimah Alomair, Dana Aldossary, Manar Alahmari, Munira Alhumaidan, Shoog Alrassan, Atta Rahman, Mustafa Youldash, Gohar Zaman, A Deep-Learning Approach to Driver Drowsiness Detection, Safety 9/2023
- Dongjiang Wu, Improving Automatic Detection of Driver Fatigue and Distraction using Machine Learning, arXiv 2024
- Y. Luo, A Review of Techniques and Methods for Deep Learning Techniques in Driver Fatigue Detection, Applied and Computational Engineering 31/2024